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Supplementary information

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Supplementary Information

Dendrite Initiation and Propagation in Lithium Metal Solid-State Batteries

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Initiation model

Initiation of a dendritic crack is considered through the pressurisation of a spherical sub-surface pore of radius, R, connected to the plated Li electrode interface via a cylindrical microcrack of length, L, and radius, a, as shown in Figure S1. The dimensions of the pore and microcrack were determined from the combination of XCT (Fig. 1) and FIB-SEM (Fig. S10). The pore radius is R = 1.5 µm. The length of the microcrack (and hence sub-surface depth of the pore) is L = 5 µm. Pores closer to the surface are unlikely to build sufficient pressure as the channel length is short and those deeper into the SE will take longer to fill. The microcrack cylinder radius is a = 0.3 µm. We do not confine our views of these cracks to be of a specific type, e.g., Griffith flaws. The distance between the two electrodes is w = 1 mm (equivalent to the thickness of the Li_6PS_5Cl pellets studied). The pore and microcrack are both assumed to be filled with lithium, therefore the system contains two domains, solid electrolyte (Domain 1) and lithium metal (Domain 2).

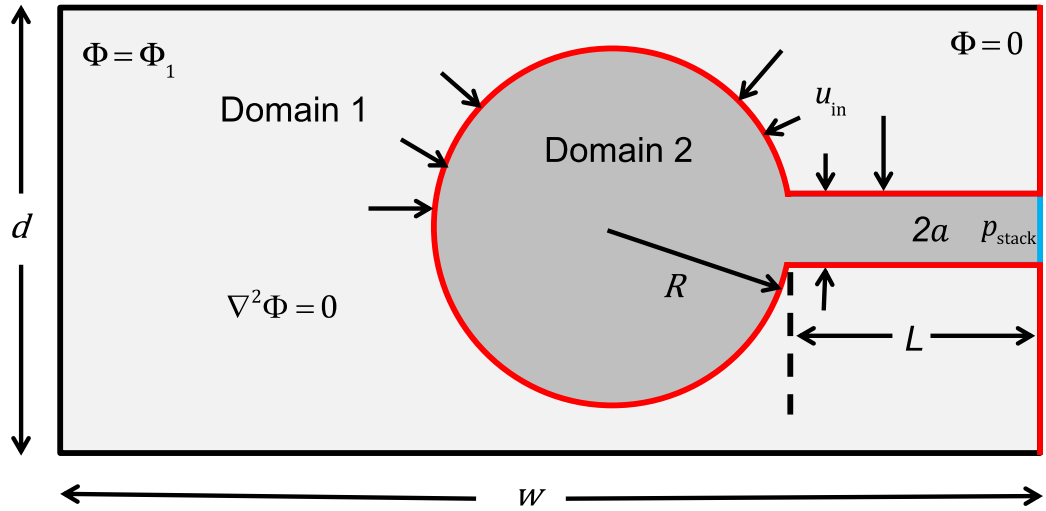


Figure S1. Dendrite initiation at a pore connected to the interface modelled with cylindrical coordinates.

For a plating current density, i, the electric potential, Φ, and electric field distribution, E, in Domain 1 can be solved from Laplace's equation, $\nabla^2\Phi = 0$, using a combination of Dirichlet and Neumann boundary conditions (for a Li_6PS_5Cl conductivity = κ = 0.1 mS/m)[1]. The system is simulated in a cylindrical coordinate with the maximum length in the radial direction given by half of the pore-pore distance d. In the presence of many pores, one pore can only affect the electric field in its neighbouring region. Dirichlet boundary conditions are applied at all electrode/solid-electrolyte interfaces, $\Phi = 0$ holds at the right (plating) electrode and $\Phi = \Phi_1 = iL/\kappa$ at the left. A Neumann boundary condition of $\partial\Phi = 0$ is applied to the edges of Domain 1 at $d/2$ away from the centre of the pore / pipe defect, hence the pore separation distance is equivalent to d which is determined to be d = 10 µm from XCT analysis, see Methods and Fig. S11. The current distribution at the lithium-electrolyte interface determined from the electric field distribution in turn determines the local deposition rate of lithium.

The plastic creep of lithium is modelled in Domain 2. The lithium is assumed incompressible

$$\rho\vec{\nabla}\cdot\vec{u} = 0 \qquad \text{(S1)}$$

The momentum balance of the lithium follows

$$\rho(\vec{u}\cdot\vec{\nabla})\vec{u} = \vec{\nabla}\cdot\left(-p\vec{\vec{I}} + \vec{\vec{\tau}}\right) \quad \text{(S2)}$$

The deformation stress τ relates to the plastic deformation-rate tensor

$$\dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}} = \frac{1}{2}\left(\vec{\nabla}\vec{u} + \vec{\nabla}\vec{u}^{T}\right) \text{ via } \vec{\vec{\tau}} = 2\eta\dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}} \quad \text{(S3)}$$

The pseudo-viscosity of lithium creep η follows a power-law plastic model

$$\eta = K\left(\frac{\dot{\bar{\varepsilon}}}{\dot{\bar{\varepsilon}}_{\mathrm{ref}}}\right)^{n-1} \quad \text{(S4)}$$

where the shear rate is $\dot{\bar{\varepsilon}} = \sqrt{2\dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}} : \dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}}}$, n is the flow behaviour index. $\dot{\bar{\varepsilon}}_{\mathrm{ref}}$ is a reference shear rate, and K is the effective viscocity at the reference shear rate $\dot{\bar{\varepsilon}}_{\mathrm{ref}}$. For lithium, $n = 0.1524$ according to Maisa et al.[2]. The reference shear rate is selected to be 2.828×10^{-7} s^{-1} and $K = 10^{12}$ Pa·s as reported by Maisa et al.[2]. In the simulation, a lower cut-off shear rate limit of 10^{-8} s^{-1} is used numerically.

As shown by Figure S1, lithium will be injected into Domain 1 via the electrochemical reaction across the surface of the pore and tube. The local velocity of lithium, u, across the lithium-electrolyte interface (perpendicularly) is related to the electric field in the solid electrolyte via

$$u_{\mathrm{in}} = \frac{\kappa M_{\mathrm{Li}}}{F\rho_{\mathrm{Li}}}E_{\perp}, \quad \text{(S5)}$$

where F = 96485 C/mol is the Faraday constant, the density and the molar mass of lithium are ρ_{Li}= 534 kg/m^3 and M_{Li} = 7 g/mol, respectively. Lithium plated in the pore and tube (red edges in Figure S1) will extrude out of the tube into the bulk anode. A constant pressure boundary condition p_{stack} is imposed on the right end of the tube, i.e., the blue edge. The velocity and stress distribution in Domain 2 can then be solved using COMSOL. The maximum hoop stress at the pore surface is equivalent to the pressure at the lithium-electrolyte interface before fracture (red boundaries), and as the pore is the largest microstructural defect in the system considered, this pressure is equivalent to the fracture strength of a Li_6PS_5Cl grain boundary as determined by microcantilever bending (see section Local Fracture Strength – Microcantilever Bending).

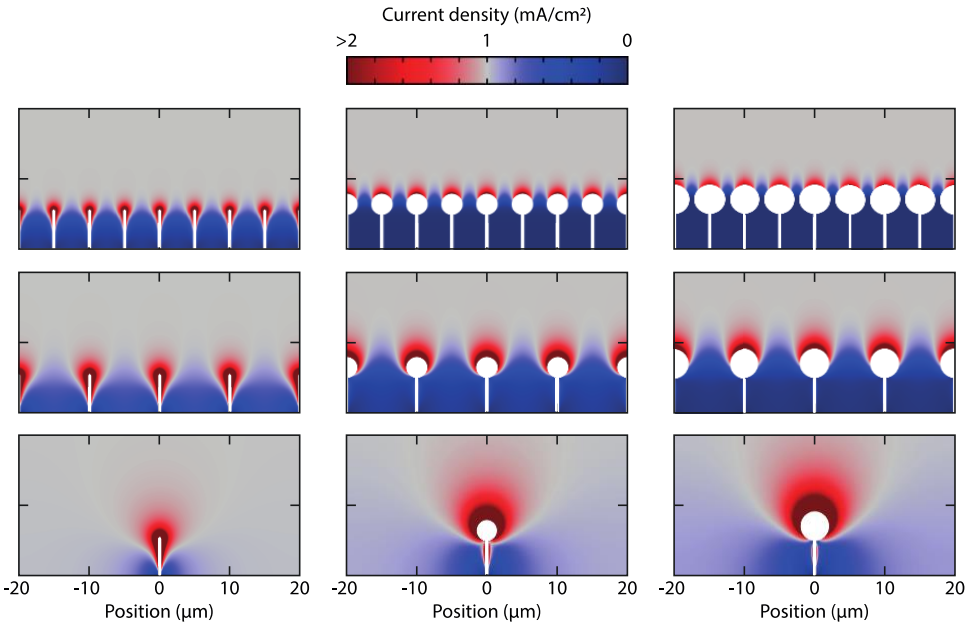


Figure S2. Current density distribution for different pore sizes 0.3 μm (left column), 1.5 μm (middle column), and 2.1 μm (right column) and different pore separations. 5 μm (top row), 10 μm (middle row), and 40 μm (bottom row). The applied current is 1 mA/cm^2. Without field enhancement and for pores that are far apart, i_{surf} equals the applied current density, and i_{pore} is lowered by the ratio of the interior pore surface area to that of the microcrack aperture at the surface of the anode. With

field enhancement, and still assuming pores are far apart, i_{pore} rises above this expected value and i_{surf} falls, as demanded by charge continuity; when pores are close together, i_{surf} falls to nearly zero and i_{pore} spreads the current across the hemispherical tips of the pores.

Propagation model

It is observed that the transverse crack propagates before the complete filling of lithium, hence a partially filled crack is considered. The initial length of the propagating dry crack is taken from XCT results (Figure 1b(iv)) which demonstrate also that it is a thin planar crack (shown in XCT 3D rendering, Fig S12). An initial crack length of l_{c}= 0.2 mm is considered, partially filled with a lithium dendrite of length l_d. The crack width is 2a = 2 µm, with a hemispherical tip. The length of the system modelled is w = 1 mm. The half width of the system is d = 2 mm.

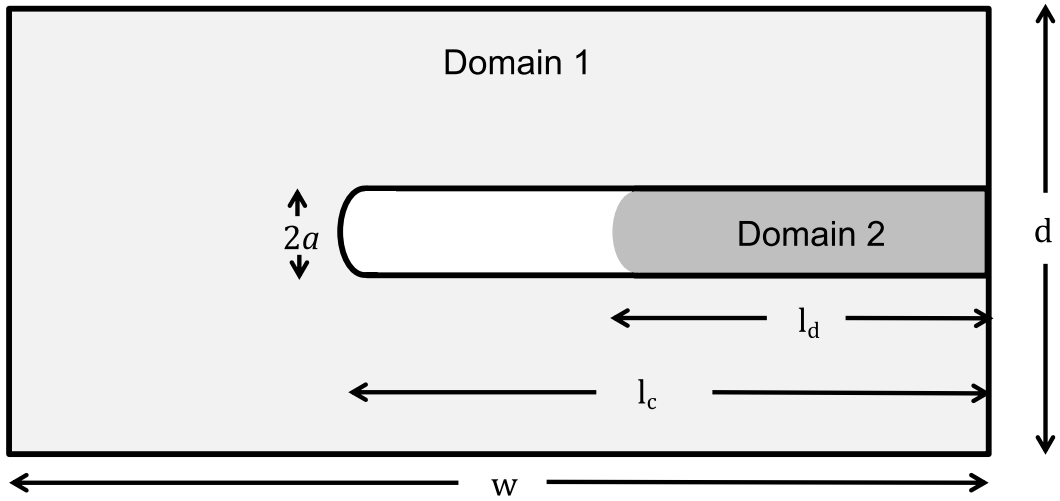


Figure S3. Crack propagation and dendrite growth

As in the initiation model, for a plating current density, i, the electric potential, Φ, and electric field distribution, E, in Domain 1 can be solved from Laplace's equation using a combination of Dirichlet and Neumann boundary conditions as shown in Figure S4. Dirichlet boundary conditions are applied at all electrode/solid-electrolyte interfaces, $\Phi = 0$ holds at the left (plating) electrode and $\Phi = \Phi_1 = iL/\kappa$ at the right. A Neumann boundary condition of $\partial\Phi = 0$ is applied to the edge of Domain 1. The current distribution at the lithium-electrolyte interface determined from the electric field distribution in turn determines the local deposition rate of lithium.

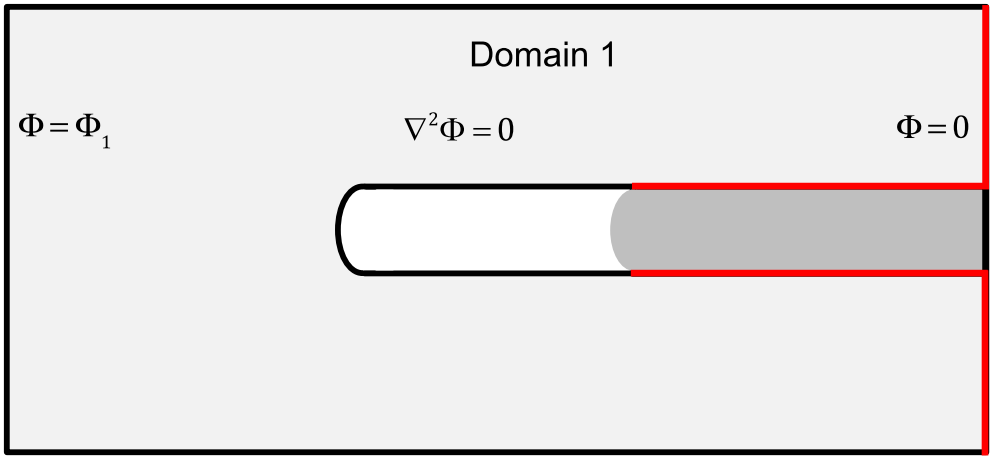


Figure S4. Model of the electric field in the solid electrolyte

Similarly, to the initiation model, plastic creep of lithium is modelled in Domain 2. As shown by Figure S4, lithium is injected into Domain 2 via the electrochemical reaction at the red boundaries. The local velocity of lithium, u, at these two boundaries is given by Eq. S5. Lithium plated on the side of the dendrite (red edges in Figure S5) will creep into the open crack from the dendrite tip or extrude out of the crack into the bulk Li anode. Two constant pressure boundary conditions are implemented on the blue edge of the domain. On the right p_{stack} is set equivalent to the applied stack pressure, on the left a tip pressure $\mathrm{p_{tip}}$ is determined from

$$p_{\mathrm{tip}} = \frac{\gamma}{r}, \qquad \text{(S6)}$$

where the lithium surface energy is γ = 0.45 J/m^2 [3]. The radius of curvature at the tip is $r = \mathrm{a}/\cos\theta$, where a is the half width of the crack and the contact angle is $\theta = \arccos(2/3) = 151°$. The velocity and stress distribution in Domain 2 can then be solved.

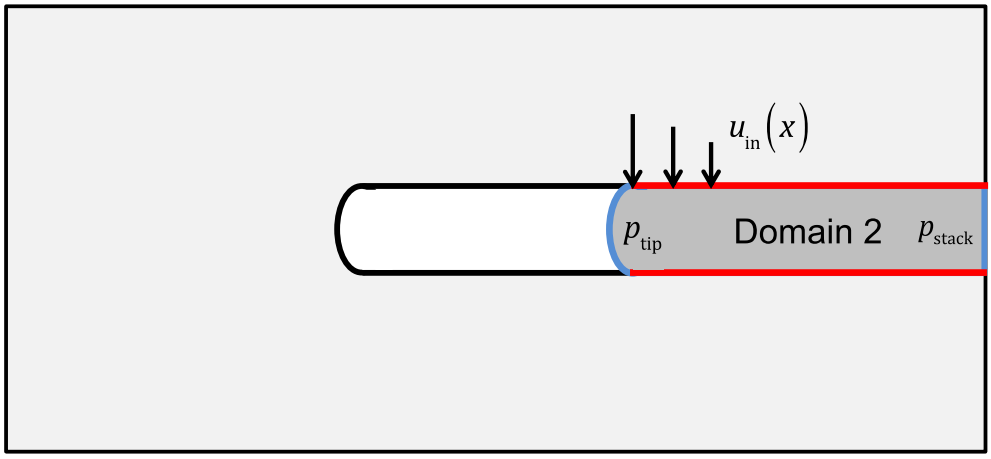


Figure S5. Model of the creeping flow in the lithium dendrite

The creeping flow in the dendrite will create a hydrodynamic stress in the lithium, which will apply a perpendicular force on the lithium-electrolyte interface as shown at the blue boundaries in Figure S6.

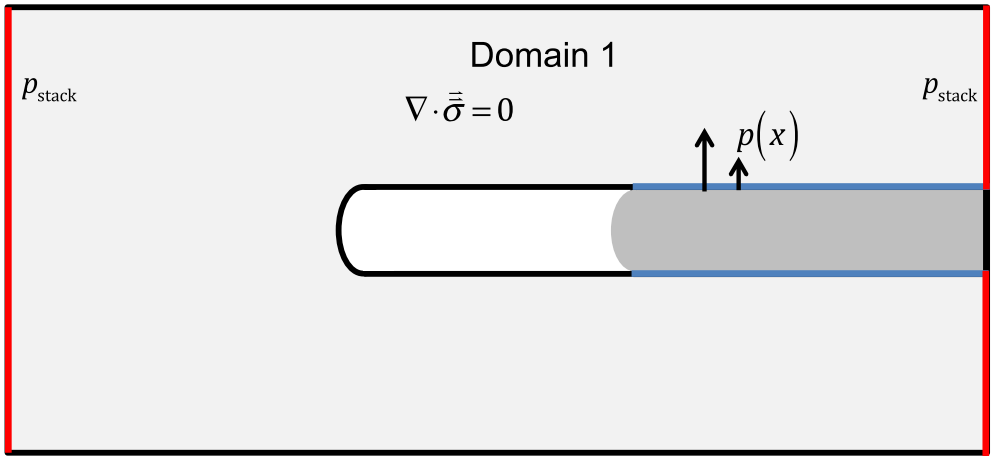


Figure S6. Model of the stress field in the solid electrolyte.

The stress field $\bar{\bar{\sigma}}$ in the solid electrolyte can be solved from the Cauchy equation $\nabla \cdot \bar{\bar{\sigma}} = 0$ after assuming linear elastic behaviour. Two pressure boundary condition $\bar{\bar{\sigma}} \cdot \hat{n} = -p_{\mathrm{stack}}$ are applied to the red boundaries, where $\hat{n}$ is the normal direction of the boundary. The force on the dendrite sides (blue boundaries) is given by the equations $\bar{\bar{\sigma}}_{\mathrm{SE}} \cdot \hat{n} = -p_{\mathrm{Li}} + \bar{\bar{\tau}}_{\mathrm{Li}} \cdot \hat{n}$ and $\left(\bar{\bar{\sigma}}_{\mathrm{SE}}\right)_{\mathrm{xy}} = 0$, where $\bar{\bar{\sigma}}_{\mathrm{SE}}$ is

the stress tensor in the solid electrolyte and $\vec{\vec{\tau}}_{\mathrm{Li}}$ is the viscous stress tensor in the lithium, $\hat{n}$ is the normal unit vector at the lithium-electrolyte interface. The other edges of Domain 1 are free.

A mode I stress intensity factor (SIF) at the dendrite tip can be calculated corresponding to the stress field in the electrolyte, with the assumption of linear elasticity. When the SIF is greater than the fracture toughness, the crack will propagate through mode I fracture. The fracture toughness of Li_6PS_5Cl is measured through nanoindentation to be 0.69±0.12 MPa·m$^{1/2}$ (see Table S3). The SIF is calculated from the energy dissipation rate at the tip (denoted by J) follows an expression of J-integral

$$\mathrm{SIF} = \sqrt{\frac{JE}{1-v^2}}, \qquad \text{(S6)}$$

where E = 28 GPa is the Young's modulus (as determined from nanoindentation, see Table S3 and Methods) and v = 0.37 is the Poisson's ratio of the solid electrolyte Li_6PS_5Cl as reported by Deng et al. [4].

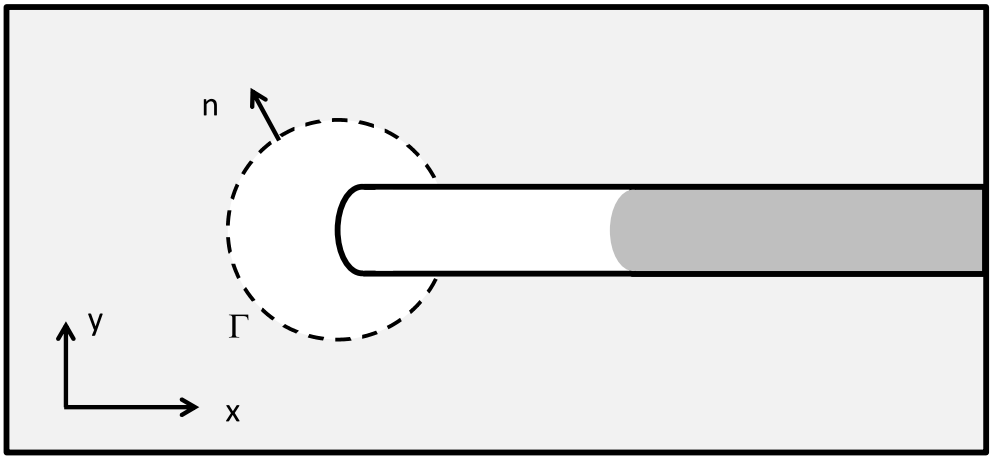


Figure S7. Calculation of J-integral.

As shown in Figure S7, if a crack opens along the y direction and propagates along the x direction, the J-integral is expressed as

$$J = \int_{\Gamma}\left[W(\mathrm{x},\mathrm{y})d\mathrm{y} - \vec{t}\cdot\frac{\partial\vec{u}}{\partial \mathrm{x}}ds\right] \qquad \text{(S7)}$$

where Γ is the integral path, $\vec{t} = \vec{\vec{\sigma}}\cdot\hat{n}$ is the traction to the outer normal of the integral path, and W is the strain energy density

$$W = W(\mathrm{x},\mathrm{y}) = \int_0^{\vec{\vec{\epsilon}}} \sigma_{\mathrm{ij}}\mathrm{d}\epsilon_{\mathrm{ij}}\ . \qquad \text{(S8)}$$

Fundamental assumptions of the models

1. Materials are assumed to be at steady states.
2. The lithium electrode is modelled as a perfect conductor and is equipotential.
3. The interface of lithium and solid electrolyte is an equipotential surface.
4. The current in the solid electrolyte is assumed to be steady, so the electric field can be solved from a Laplace's equation.
5. The plastic creep of lithium is modelled as a viscous flow.
6. The creep velocity of lithium at the interface of lithium and solid electrolyte is determined by the speed of lithium plating/stripping calculated from the interfacial current (eq. S5).
7. The solid electrolyte is assumed to has a linear elasticity.

The governing equations are summarized in Table S1 and S2.

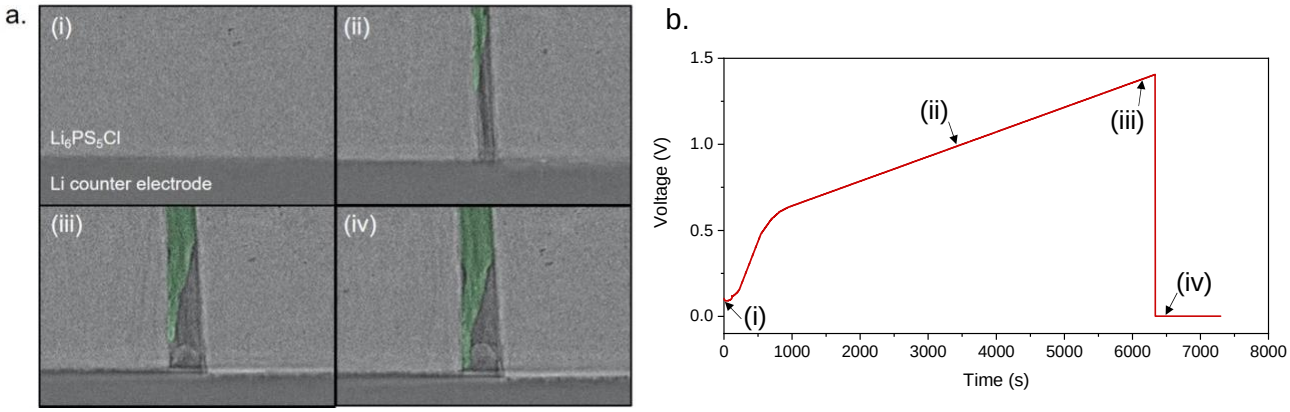


Figure S8: (a) *Operando* XCT images showing the growth of Li metal within the crack which has reached the counter electrode. Li metal propagating within the crack from the working electrode towards the counter electrode is highlighted in green. (b) Voltage profile of the corresponding Li/Li_6PS_5Cl/Li cell during the single plating. The drop in voltage indicates the short-circuit and occurs at the point when the Li metal reaches the counter electrode from within the crack.

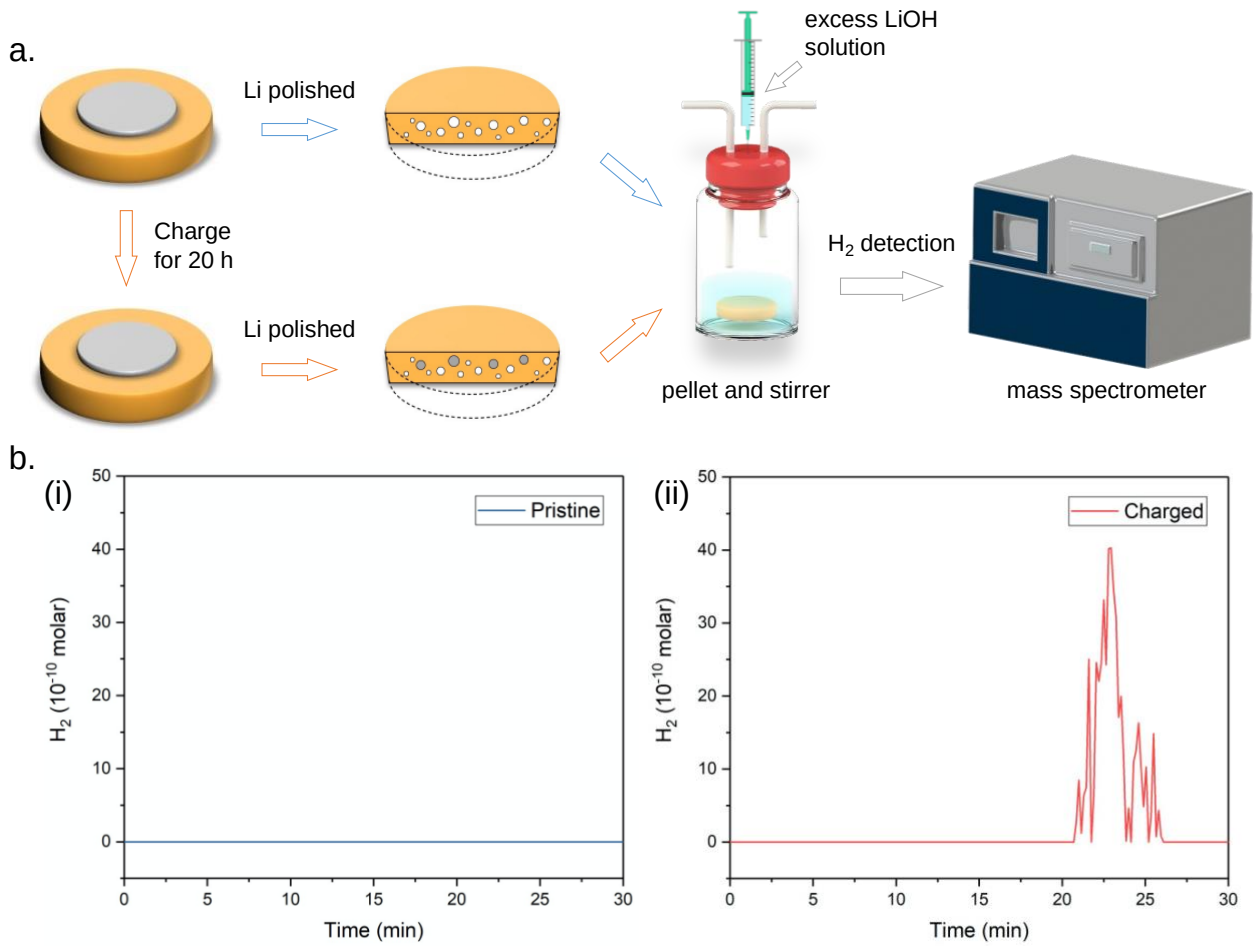


Fig. S9. The Li anode is removed then the SE subjected to slow etching/dissolution while the gas is monitored by mass spectrometry. No Li is detected in pristine sample whereas there is detection of sub-surface Li deposition in the sample after charging at 0.05 mA/cm^2 for 2 mAh/cm^2, indicating the presence of Li in sub-surface regions of SE after charging.

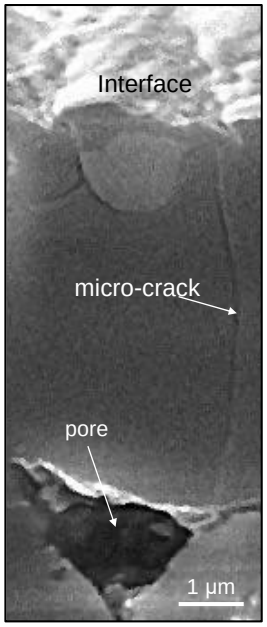


Figure S10. FIB-SEM of Li_6PS_5Cl pellet, showing the presence of a sub-surface pore connected to the interface via a micro-crack.

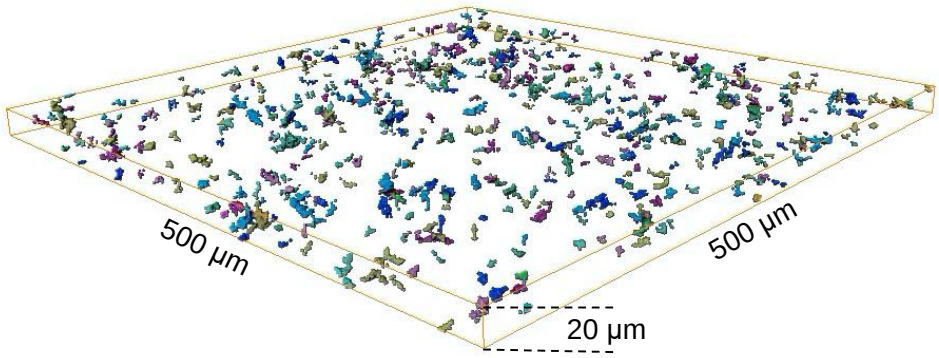


Fig. S11. 3D rendering of pores in a 500 × 500 × 20 μm^3 sub-volume of the solid electrolyte adjacent to the plated interface from the *Operando* XCT scan at pristine in Fig. 1. Colours are used to distinguish the neighboured pores. The sub-volume was selected for the quantitative analysis of pore distribution in Avizo Fire. Pores larger than 2 × 2 × 2 voxels (3.25 × 3.25 × 3.25 μm^3) in the sub-volume were segmented out with interactive thresholding. The distance between pores in the X-Y plane (parallel to the Li/Li_6PS_5Cl interface) were calculated based on the space coordinates of pores. The average distance from a pore to a nearest neighboured pore was then calculated and used as the average distance between pores in the dendrite initiation modelling.

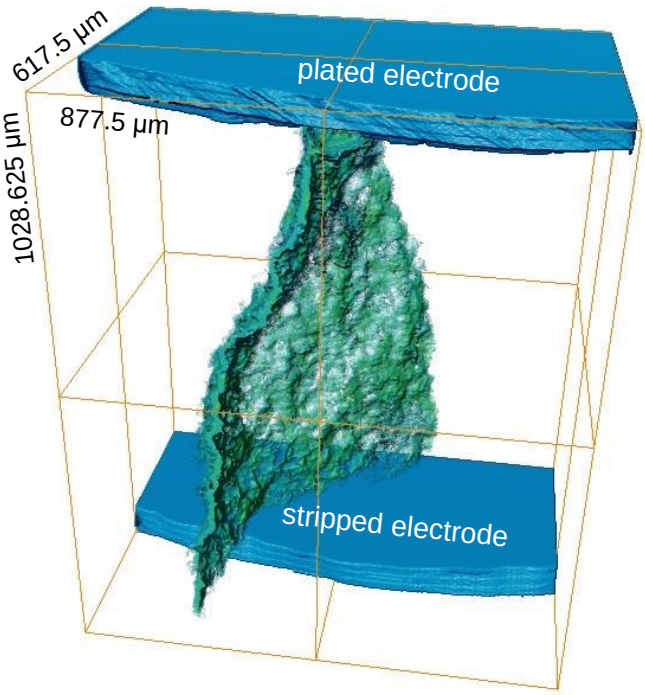


Fig. S12. 3D rendering of the dendritic crack and Li electrode from the Operando XCT scan of Fig. 1b(vii).

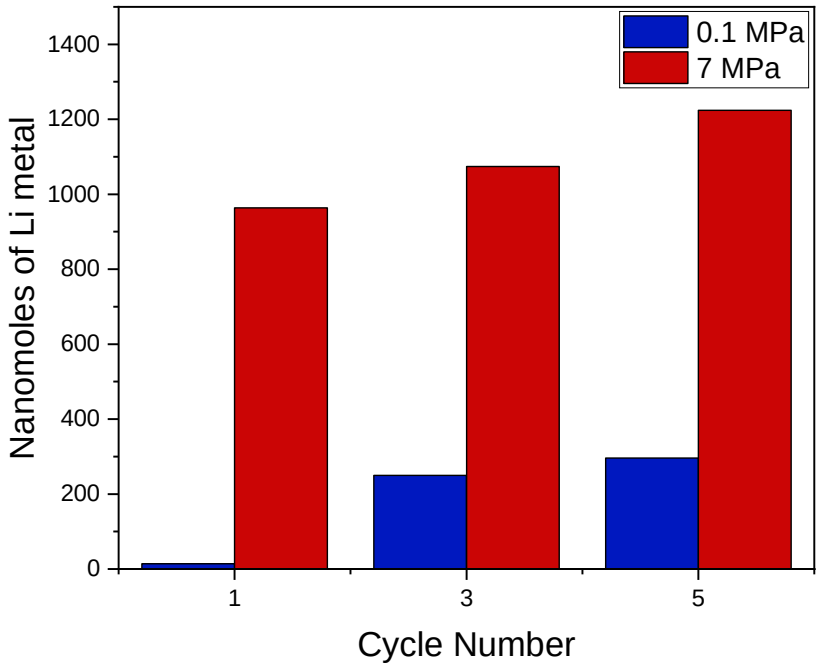


Figure S13: Quantification of Li metal within the solid electrolyte after stripping, determined through mass spectrometry. The Li anode is removed then the solid electrolyte subjected to slow etching/dissolution while the gas is monitored by mass spectrometry. Cells are cycled under 4 mA/cm^2 plating current and 0.05 mA/cm^2 stripping current for 0.5 mAh/cm^2 each half cycle, in accord with the cells in Figure 4. The amount of Li metal in the solid electrolyte cycled only once at 0.1 MPa was undetectable. The results demonstrate the effect of pressure in encouraging dendrite formation in solid electrolyte and how dendrites progressively grow over multiple cycles.

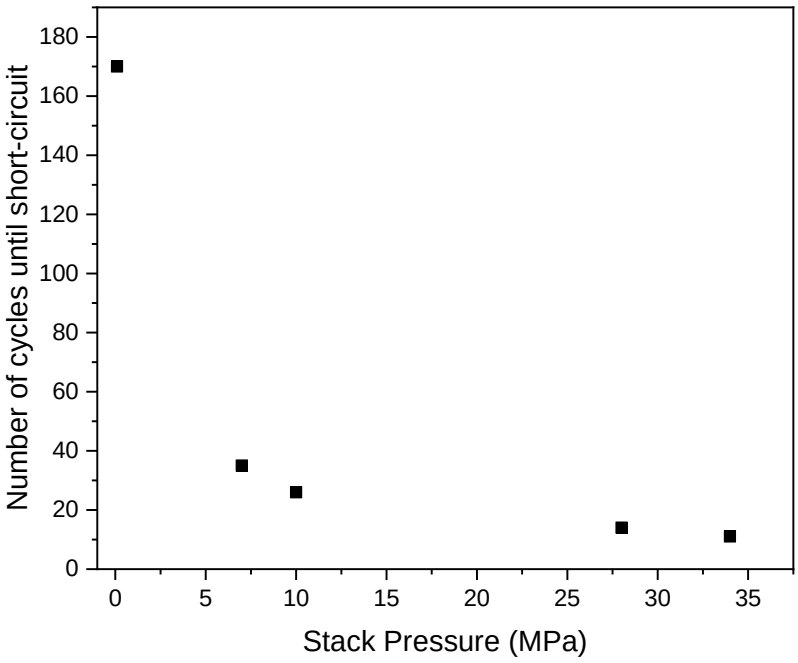


Figure S14: Number of cycles to short-circuit vs applied stack pressure for Li/Li_6PS_5Cl 3-electrode cells cycled under identical conditions to Figure 4 (4 mA.cm^{-2} plating current density, 0.05 mA.cm^{-2} stripping current density). Cells cycled at 10MPa, 28MPa and 34MPa failed at the 26th, 14th and 11th cycle respectively.

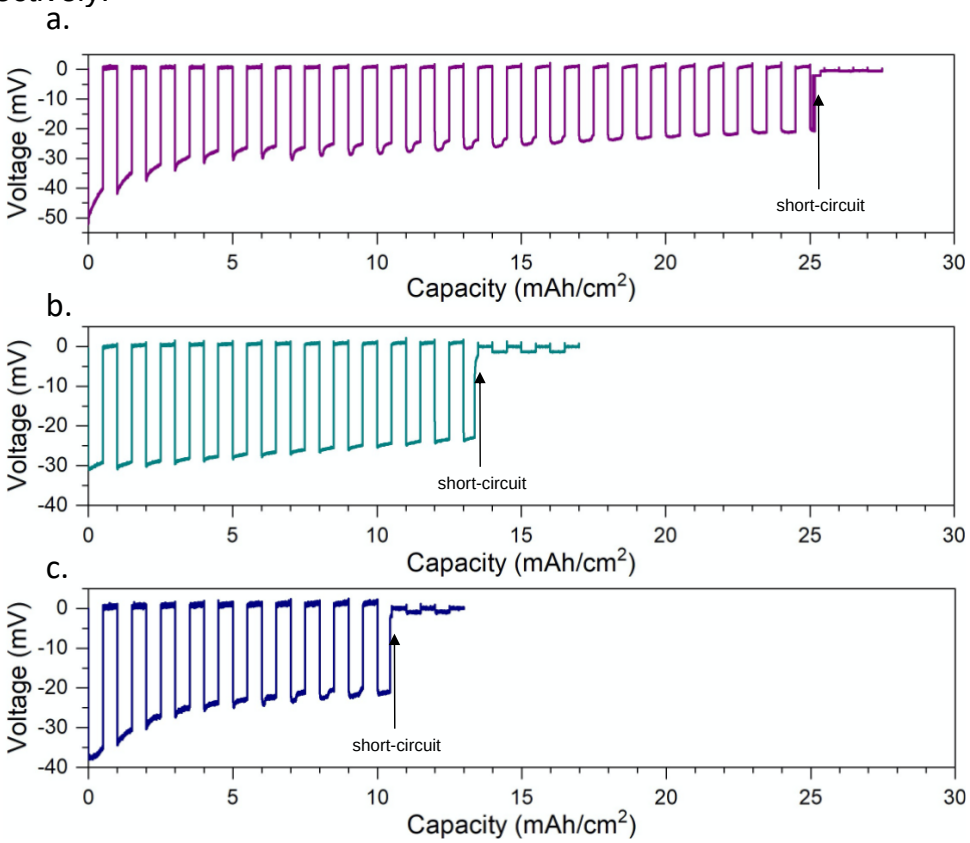


Figure S15: Voltage profiles corresponding to data points at (a) 10 MPa, (b) 28 MPa and (c) 34 MPa in Figure S14. The voltage profiles for the other two data points in Figure S14 are in Figure 4. Cycling of employed Li/Li_6PS_5Cl 3-electrode cells with current densities of 4.0 mA/cm^2 for plating, 0.05 mA/cm^2 for stripping (to avoid void formation), both for a capacity of 0.5 mAh/cm^2.

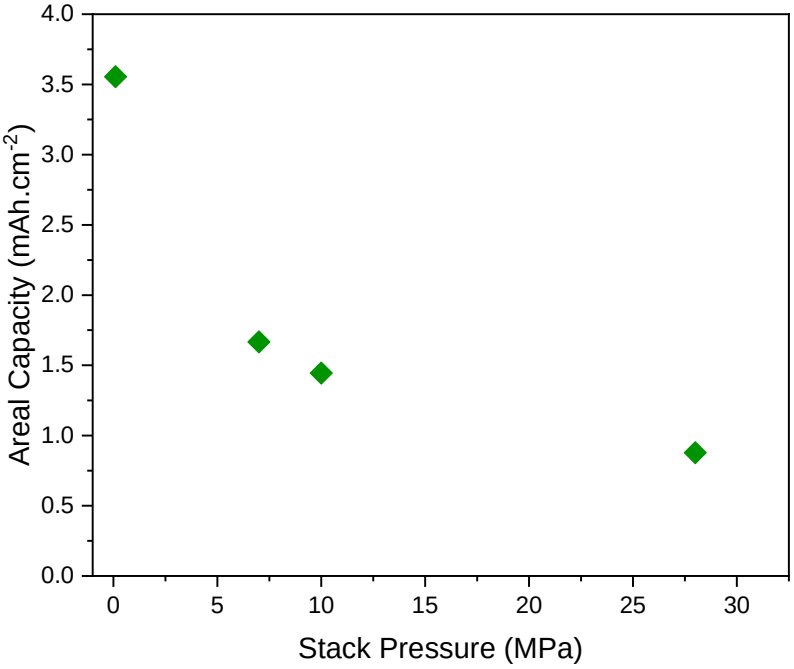


Figure S16: Capacity required to induce short-circuit versus stack pressure for Li/Li_6PS_5Cl cells subject to single plating at 4 $mA.cm^{-2}$ current density. The cells are different from those reported in Figure S14. Cells under 0.1 MPa (ambient pressure), 7 MPa, 10 MPa and 28 MPa short-circuited after plating 3.6 $mAh.cm^{-2}$, 1.7 $mAh.cm^{-2}$, 1.4 $mAh.cm^{-2}$ and 0.9 $mAh.cm^{-2}$ respectively.

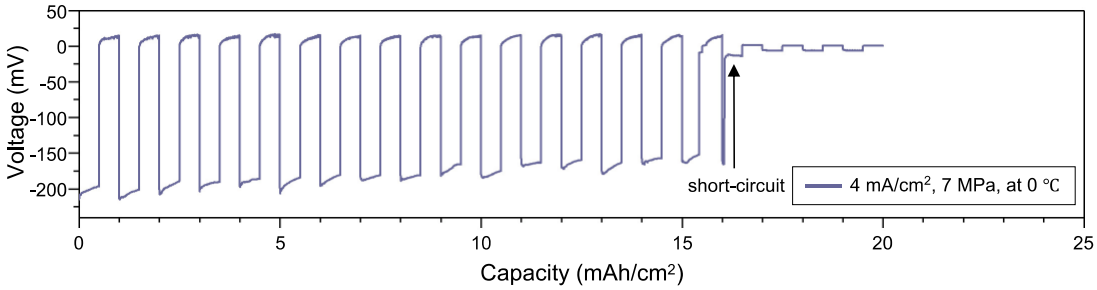


Figure S17: Cycling of Li/Li_6PS_5Cl 3-electrode cell with current densities of 4.0 mA/cm^2 for plating, 0.05 mA/cm^2 for stripping, both for a capacity of 0.5 mAh/cm^2, under 7 MPa pressure at 0°C. Failure occurs on the 16th cycle, which is earlier than in Figure 4d (35 cycles), which was cycled at 30°C, this is due to the limited creep of Li at low temperature.

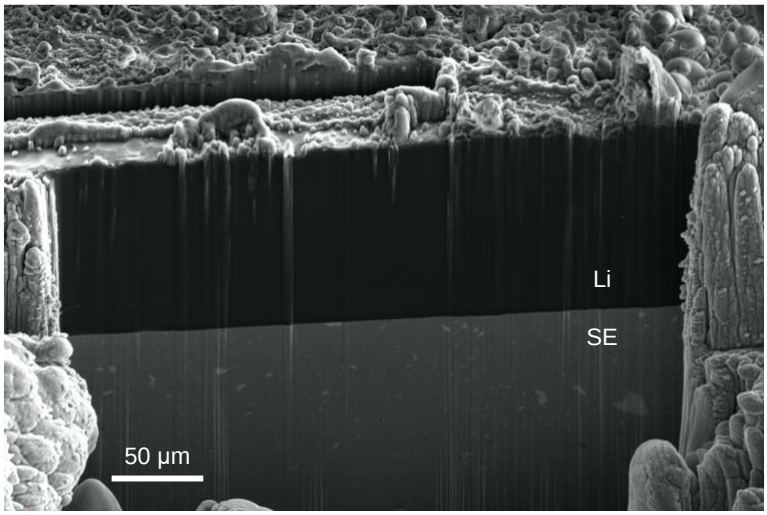


Figure S18: FIB-SEM images demonstrating good conformal contact between the Li metal electrode and Li_6PS_5Cl solid electrolyte.

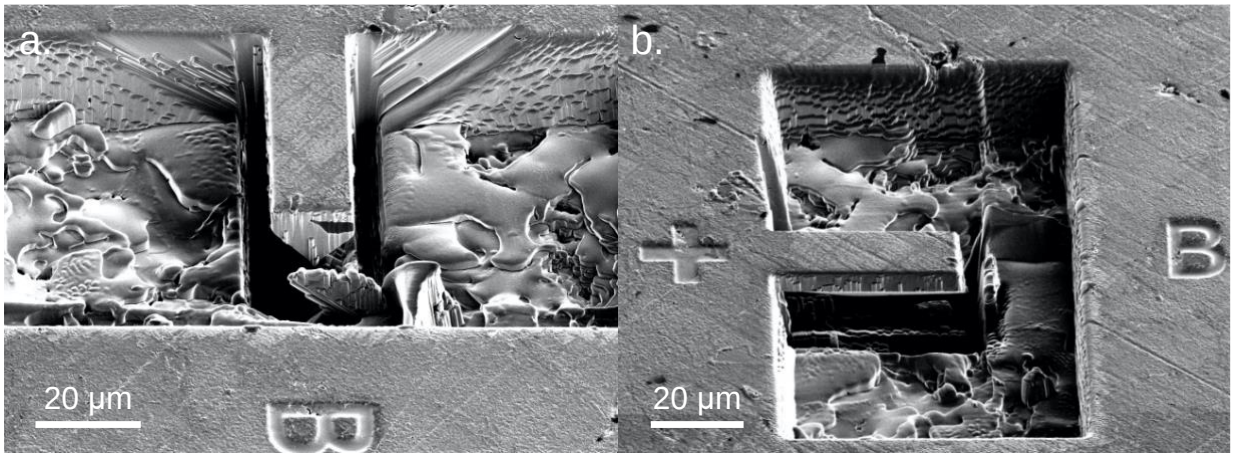


Figure S19. (a) SEM image of a microcantilever of Li_6PS_5Cl prepared using PFIB. (b) Side view SEM images showing "+" reference point.

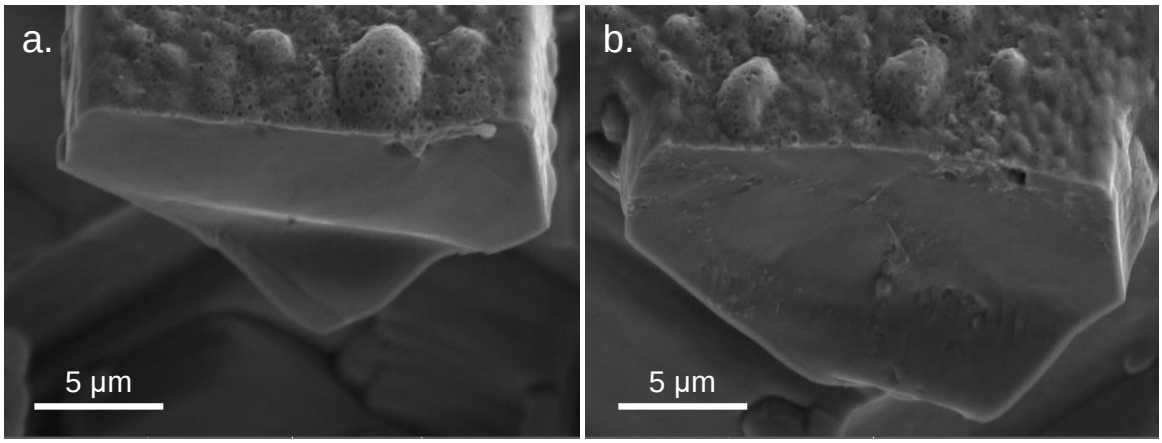


Figure S20: SEM images of the fracture surfaces of Li_6PS_5Cl microcantilevers.

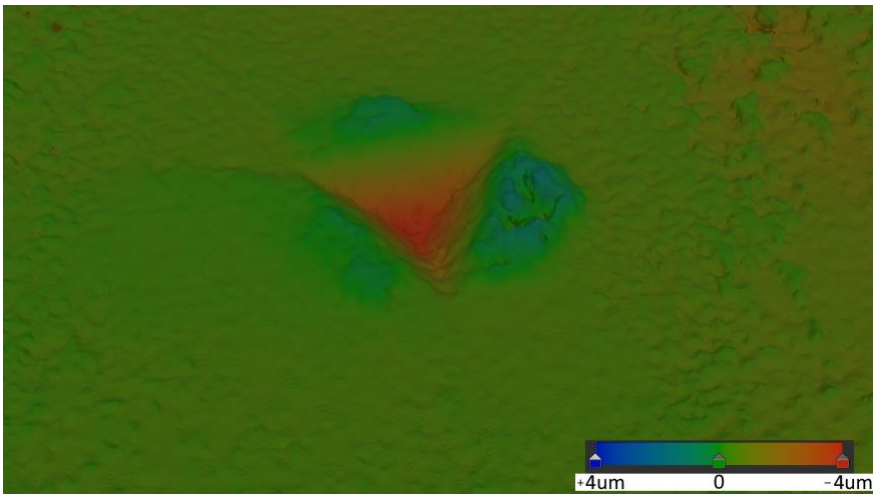


Figure S21: 3D reconstructed height map of an Li_6PS_5Cl indent.

Table S1. Governing equations and boundary conditions in the initiation model

Region	Equations	
Lithium bulk	$\rho\vec{\nabla}\cdot\vec{u}=0$	(S1)
	$\rho(\vec{u}\cdot\vec{\nabla})\vec{u}=\vec{\nabla}\cdot\left(-p\vec{\vec{I}}+\vec{\vec{\tau}}\right)$	(S2)
	$\dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}}=\frac{1}{2}\left(\vec{\nabla}\vec{u}+\vec{\nabla}\vec{u}^{T}\right),\quad \vec{\vec{\tau}}=2\eta\dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}}$	(S3)
	$\eta=K\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{\mathrm{ref}}}\right)^{n-1},\quad \dot{\varepsilon}=\sqrt{2\dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}}:\dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}}}$	(S4)
Boundary condition of lithium	$\left(-p\vec{\vec{I}}+\vec{\vec{\tau}}\right)\cdot\hat{n}=-p_{\mathrm{stack}}\hat{n}$ at the right (plating) boundary in Fig. S1	
Solid electrolyte bulk	$\nabla^2\Phi=0,\ \vec{E}=-\vec{\nabla}\Phi$	
Boundary condition of solid electrolytes	$\Phi=0$ at the right (plating) boundary in Fig. S1	
	$\Phi=\Phi_1=iL/\kappa$ at the left (stripping) boundary in Fig. S1	
	$\partial\Phi=0$ at the top and bottom boundaries in Fig. S1	
Lithium-electrolyte Interface	$\Phi=0$	
	$u_{\mathrm{in}}=\frac{\kappa M_{\mathrm{Li}}}{F\rho_{\mathrm{Li}}}E_{\perp}$	(S5)

Table S2. Governing equations and boundary conditions in the propagation model

Region	Equations	
Lithium bulk	$\rho\vec{\nabla}\cdot\vec{\mathrm{u}}=0$	(S1)
	$\rho(\vec{u}\cdot\vec{\nabla})\vec{u}=\vec{\nabla}\cdot\left(-p\vec{\vec{I}}+\vec{\vec{\tau}}\right)$	(S2)
	$\dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}}=\frac{1}{2}\left(\vec{\nabla}\vec{u}+\vec{\nabla}\vec{u}^{T}\right),\ \vec{\vec{\tau}}=2\eta\dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}}$	(S3)
	$\eta=K\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{\mathrm{ref}}}\right)^{n-1},\quad \dot{\varepsilon}=\sqrt{2\dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}}:\dot{\vec{\vec{\varepsilon}}}_{\mathrm{p}}}$	(S4)
Boundary condition of lithium	$\left(-p\vec{\vec{I}}+\vec{\vec{\tau}}\right)\cdot\hat{n}=-p_{\mathrm{stack}}\hat{n}$ at the right (plating) boundary in Fig. S5	
	$\left(-p\vec{\vec{I}}+\vec{\vec{\tau}}\right)\cdot\hat{n}=-p_{\mathrm{tip}}\hat{n},\ p_{\mathrm{tip}}=\frac{\gamma}{r}$ at the left (tip) boundary	
Solid electrolyte bulk	$\nabla^2\Phi=0,\ \vec{E}=-\vec{\nabla}\Phi$	
	$\nabla\cdot\vec{\vec{\sigma}}_{\mathrm{SE}}=0,\ \vec{\vec{\sigma}}_{\mathrm{SE}}=-p_{\mathrm{SE}}\vec{\vec{I}}+\vec{\vec{\tau}}_{\mathrm{SE}}$	
	Linear elasticity	
Boundary condition of solid electrolytes	$\Phi=0$ at the right (plating) boundary in Fig. S4	
	$\Phi=\Phi_1=iL/\kappa$ at the left (stripping) boundary in Fig. S4	
	$\partial\Phi=0$ at the other boundaries in Fig. S4	
	$\vec{\vec{\sigma}}\cdot\hat{n}=-p_{\mathrm{stack}}\hat{n}$ at the left and right boundaries in Fig. S6	

	$\vec{\vec{\sigma}} \cdot \hat{n} = 0$ at the other boundaries in Fig. S6	
Lithium-electrolyte Interface	$\Phi = 0$	
	$\vec{u} \cdot \hat{n} = \frac{\kappa M_{\mathrm{Li}}}{F \rho_{\mathrm{Li}}} \vec{E} \cdot \hat{n}$	(S5)
	$\vec{\vec{\sigma}}_{\mathrm{SE}} \cdot \hat{n} = -p_{\mathrm{Li}} + \vec{\vec{\tau}}_{\mathrm{Li}} \cdot \hat{n}$	

Table S3. List of Li_6PS_5Cl mechanical properties determined in this work.

Name	Symbol	Unit	Method	Value
Elastic Modulus	E	GPa	Nanoindentation	28.0 ± 1.8
Hardness	H	MPa	Nanoindentation	921 ± 196
Fracture Toughness	K_{IC}	MPa $m^{1/2}$	Nanoindentation	0.69 ± 0.12
Local Fracture Strength	σ_c	MPa	Microcantilever bending	91.1 ± 14.5

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